Run three machine cycles

Name ____________________________________

Your job: run three instructions. Each one goes through the same four stages — **FETCH → DECODE → EXECUTE → WRITE**. The instructions are shortened codes, so you have to look each one up in the table to find out what it means.

MEMORY

`0: LDI 5`
`1: ADI 1`
`2: STA 20`
`20: (data — old value not needed)`

STARTING STATE

`PC = 0`
`IR = — (none fetched)`
`A = ? (not needed)`
`M[20] = ? (old value)`

DECODE LOOKUP TABLE

Code	What it means	Destination
LDI n	Load the number n into register A	A
ADI n	Add the number n to what is in A	A
SBI n	Subtract the number n from what is in A	A
STA n	Copy what is in A into memory address n	M[n]

The trace — one row per instruction, one column per stage

Instruction	**1. FETCH** Copy the instruction at the PC into the IR, then move the PC on.	**2. DECODE** Look the code up. What does it mean, and where does the result go?	**3. EXECUTE** Work out the result. Nothing is stored yet.	**4. WRITE** Put the result in the destination you decoded.
Cycle 1 `PC starts at 0`	IR holds ______ PC is now ______	Means ______ Destination ______	Result ______	A = ______
Cycle 2 `use the PC you wrote`	IR holds ______ PC is now ______	Means ______ Destination ______	Calculation ______ Result ______	A = ______
Cycle 3 `use the PC you wrote`	IR holds ______ PC is now ______	Means ______ Destination ______	Value to be copied ______	Destination and value ______

Stuck on a box? Every box has its own hint on the Canvas page for this activity.

Final check

Ending state: PC = ________ A = ________ M[20] = ________

You ran **three** instructions. How many stages did each one pass through? ________

In cycle 3 the result did not go into register A. Which stage told you that?

____

Why could the machine not skip the decode stage on this worksheet?

____

Model boundary: a real machine stores instructions as numbers, not as short codes like **LDI**, and hardware implements the decoding rules. We hide that encoding so the cycle stays visible. **?** means “not needed for this trace,” not physically empty.